

A bp-continuous holomorphic functional that is not compact-open continuous

An explicit example requested in arXiv:2403.11712

Autonomous solve-first investigation — lane 12

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Abstract

Haak–Haase ask for an example, on an open subset of an arbitrary Banach space, of a bp-continuous functional on H^∞ which is not continuous on bounded sets for the compact-open topology τ_c . Let $X = C([0, \omega_1])$ and $O = \mathbb{B}_X$. For $f \in H^\infty(O)$, the derivative $Df(0)$ is a Radon measure on $[0, \omega_1]$; set

$$\Phi(f) = Df(0)(\{\omega_1\}).$$

Every sequence of such measures has jointly countable support, which makes Φ bp-continuous. In contrast, the net of linear functions $g_\beta(x) = x(\beta)$ converges to g_{ω_1} uniformly on compact subsets as $\beta \uparrow \omega_1$, while $\Phi(g_\beta) = 0$ and $\Phi(g_{\omega_1}) = 1$. Thus Φ fails τ_c -continuity even on the unit ball.

1 The source question

For an open set O in a Banach space, let $H^\infty(O)$ be the Banach space of bounded scalar holomorphic functions. A sequence converges *bounded pointwise* (bp) if it is uniformly bounded in supremum norm and converges pointwise. The topology τ_c is uniform convergence on compact subsets of O .

Remark 3.2(2) of Haak–Haase [1] extends their Linearization Theorem to an arbitrary Banach-space domain when the vector-valued holomorphic map is separably valued. They observe that a bp-continuous functional need not be τ_c -continuous on bounded sets, but explicitly state that they know no example. The construction below supplies one.

2 The sequentially continuous endpoint mass

Let $K = [0, \omega_1]$ carry the order topology, and let $X = C(K)$ over $\mathbb{C}$. Since K is scattered, every finite Radon measure on K is atomic with countable support. Thus the Riesz representation gives

$$X' = M(K) = \ell^1(K).$$

Define the norm-one functional $T : X' \rightarrow \mathbb{C}$ by

$$T\mu = \mu(\{\omega_1\}).$$

Lemma 1. *T is sequentially w^* -continuous on X' , but its restriction to the dual unit ball is not w^* -continuous.*

Proof. Suppose $\mu_n \rightarrow \mu$ in $\sigma(X', X)$. The union of the supports of $\mu, \mu_1, \mu_2, \dots$, after deleting ω_1 , is countable. Its supremum is therefore some countable ordinal $\alpha < \omega_1$. The tail $(\alpha, \omega_1]$ is clopen in K , so $u = \mathbf{1}_{(\alpha, \omega_1]}$ belongs to $C(K)$. By the choice of α ,

$$T\mu_n = \langle u, \mu_n \rangle \longrightarrow \langle u, \mu \rangle = T\mu.$$

For noncontinuity, let δ_β denote point mass at β . Every $x \in C(K)$ is constant on a sufficiently late ordinal interval. Hence $\delta_\beta \rightarrow \delta_{\omega_1}$ in $\sigma(X', X)$ as the net of countable ordinals β increases to ω_1 . Yet $T\delta_\beta = 0$ for every $\beta < \omega_1$, whereas $T\delta_{\omega_1} = 1$. $\square$

3 Transfer to bounded holomorphic functions

Take the open unit ball $O = \mathbb{B}_X$, and define

$$\Phi : H^\infty(O) \longrightarrow \mathbb{C}, \quad \Phi(f) = T(Df(0)) = Df(0)(\{\omega_1\}). \quad (1)$$

Cauchy's estimate on complex lines gives $\|Df(0)\| \leq \|f\|_\infty$, so Φ is in fact norm-continuous.

Lemma 2. *If $f_n \rightarrow f$ bounded pointwise in $H^\infty(O)$, then $Df_n(0) \rightarrow Df(0)$ in $\sigma(X', X)$.*

Proof. The derivatives are norm-bounded by Cauchy’s estimate. Fix $x \in X$ and choose $r > 0$ with $r\|x\| < 1$. The scalar holomorphic functions

$$h_n(z) = f_n(zx), \quad |z| < r,$$

are uniformly bounded and converge pointwise to $h(z) = f(zx)$. The one-variable normal-family theorem makes this convergence locally uniform; hence $h'_n(0) \rightarrow h'(0)$. Since $h'_n(0) = Df_n(0)x$, this is precisely weak-star convergence of the derivatives. $\square$

Combining Lemmas 1 and 2 proves that Φ is bp-continuous. It remains to show that the stronger-looking norm continuity does not rescue compact-open continuity on bounded sets.

For $\beta \leq \omega_1$, define

$$g_\beta(x) = x(\beta), \quad x \in \mathbb{B}_X.$$

Each g_β is a linear holomorphic function of supremum norm one. The net g_β converges to g_{ω_1} in τ_c : it converges pointwise because every $x \in C(K)$ is eventually constant, and the functionals δ_β have a common norm bound, so a finite ε -net upgrades pointwise convergence to uniform convergence on every norm-compact subset of $\mathbb{B}_X$. On the other hand,

$$\Phi(g_\beta) = T\delta_\beta = 0 \quad (\beta < \omega_1), \quad \Phi(g_{\omega_1}) = T\delta_{\omega_1} = 1.$$

Theorem 3. *The functional (1) is bp-continuous on $H^\infty(\mathbb{B}_{C([0, \omega_1])})$ but is not τ_c -continuous on its unit ball. Consequently it is the example requested in Remark 3.2(2) of arXiv:2403.11712.*

4 Checks, mechanism, and scope

The construction separates sequences from nets exactly where required. Joint countability of the supports proves sequential continuity; the uncountable ordinal net witnesses genuine topological discontinuity. No separability is assumed of the domain space. The result is even slightly stronger than requested because Φ is norm-continuous and fails τ_c -continuity already on the unit ball.

More generally, the same derivative-at-zero construction works whenever a Banach space Y admits a bounded functional on Y' which is weak-star sequentially continuous but not weak-star continuous on a bounded set—in particular, whenever Y fails the Mazur property in this form.

Exact source-phrase, arXiv-id, bp/compact-open, and holomorphic/Mazur-property searches on 13 August 2026 found the source question but no answer to it. The endpoint-mass map on $M([0, \omega_1])$ is a classical weak-star sequential counterexample; the transfer through $Df(0)$ is the solve-first step. This bounded search is not a priority claim.

Human-review recommendation. Accept as a candidate full counterexample. The two substantive checks are the jointly countable-support argument in Lemma 1 and the compact-uniform convergence of the evaluation net; both are explicit.

References

- [1] B. H. Haak and M. Haase, *Vector-Valued Holomorphic Functions and Abstract Fubini-Type Theorems*, Archiv der Mathematik **123** (2024), 275–290; [arXiv:2403.11712v2](#).